

Name: _____ **:Date** _____

Course: SCH4U

Time: 9.00am - 11.00am (2 hours)

Teacher:

Grade 12 University Preparation Chemistry Exam

Instructions:

- **Please read the first page carefully.**
 - Show all your work.
 - Answer in the space provided.
 - No electronic devices allowed (unless specified).
 - Write legibly.
 - Formulas/tables will be provided on the back of the exam.
 - Check your work before submitting.
 - Any form of cheating, plagiarism, or unauthorized collaboration will result in zero marks and possible disciplinary action.
 - If any question is unclear, raise your hand

Multiple Choice (30 minutes)

Short Answer (90 minutes)

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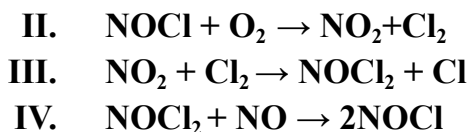
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1. Which of the following changes will **not** increase the rate of a reaction according to collision theory?
 - a. Increasing temperature
 - b. Increasing surface area of reactants
 - c. Adding a catalyst
 - d. Decreasing reactant concentration
 - e. Increasing the pressure of a solid reactant
2. Which of the following is the most polar bond
 - a. H-F
 - b. H-Br
 - c. H-Cl
 - d. H-S
 - e. H-I
3. A solution is prepared by dissolving solid ammonium nitrate, $\text{NH}_4\text{NO}_{3(s)}$, in water. The initial temperature of the water was 21.0°C , but after the solid had dissolved, the temperature had fallen to 11.5°C . What conclusions may be made about ΔH and ΔS ?
 - a. $\Delta H > 0$ and $\Delta S < 0$
 - b. $\Delta H > 0$ and $\Delta S > 0$
 - c. $\Delta H < 0$ and $\Delta S < 0$
 - d. $\Delta H < 0$ and $\Delta S > 0$
 - e. $\Delta H = 0$ and $\Delta S > 0$
4. Which of the following elements has the highest first ionization energy?
 - a. Na
 - b. C
 - c. N
 - d. P
 - e. Al
5. Which experiment demonstrated that most of an atom's positive charge is concentrated in a small, dense nucleus by observing the deflection of positively charged particles?
 - a. Cathode ray tube experiment, which studied the behavior of electrons
 - b. Oil drop experiment, which measured the charge of electrons
 - c. Gold foil experiment, which observed alpha particles being repelled at large angles
 - d. Atomic emission spectra, which showed quantized energy levels
 - e. Photoelectric effect, which demonstrated the particle nature of light
6. At 0 K, the value of which of the following is exactly zero for a perfectly ordered crystalline substance?

-
- a. Enthalpy
b. Entropy
c. Internal energy
d. Gibbs free energy
e. Heat capacity
7. Which of the following is a conjugate acid–base pair in the reaction $\text{HSO}_3^- + \text{H}_2\text{O} \rightleftharpoons \text{SO}_3^{2-} + \text{H}_3\text{O}^+$?
- a. $\text{HSO}_3^- / \text{SO}_3^{2-}$
b. $\text{HSO}_3^- / \text{H}_3\text{O}^+$
c. $\text{H}_2\text{O} / \text{SO}_3^{2-}$
d. $\text{SO}_3^{2-} / \text{H}_3\text{O}^+$
e. $\text{H}_2\text{O} / \text{HSO}_3^-$
8. Which of the following substances forms a metallic bond at STP?
- a. $\text{K}_{(\text{s})}$
b. $\text{F}_{2(\text{g})}$
c. $\text{Al}_2\text{O}_{3(\text{s})}$
d. $\text{CO}_{2(\text{g})}$
e. $\text{NaCl}_{(\text{s})}$
9. Which of the following molecules contains the carbon–carbon bond that requires the most energy to break?
- a. C_2H_6
b. CH_4
c. C_2H_2
d. C_3H_8
e. $\text{C}_2\text{H}_2\text{O}$
10. Given the galvanic cell notation:
- $$\text{Zn}_{(\text{s})} \mid \text{Zn}^{2+}_{(\text{aq})} \parallel \text{Cu}^{2+}_{(\text{aq})} \mid \text{Cu}_{(\text{s})}$$
- Which of the following correctly displays the oxidation process?
- a. $\text{Cu}^{2+}_{(\text{aq})} + 2\text{e}^- \rightarrow \text{Cu}_{(\text{s})}$
b. $\text{Zn}^{2+}_{(\text{aq})} + 2\text{e}^- \rightarrow \text{Zn}_{(\text{s})}$
c. $\text{Zn}_{(\text{s})} \rightarrow \text{Zn}^{2+}_{(\text{aq})} + 2\text{e}^-$
d. $\text{Cu}_{(\text{s})} \rightarrow \text{Cu}^{2+}_{(\text{aq})} + 2\text{e}^-$
e. Electrons flow from $\text{Cu}_{(\text{s})}$ to $\text{Zn}_{(\text{s})}$
11. Solid NaOH produces a higher pH in solution than solid $\text{Ca}(\text{OH})_2$ under similar conditions. Why?

-
- a. NaOH undergoes hydrolysis in water while $\text{Ca}(\text{OH})_2$ does not
- b. $\text{Ca}(\text{OH})_2$ has a lower ionization energy than NaOH
- c. NaOH undergoes greater ionization in water
- d. $\text{Ca}(\text{OH})_2$ is insoluble in water
- e. $\text{Ca}(\text{OH})_2$ forms a weak base in solution
12. Which of the following processes does NOT involve oxidation or reduction?
- a. $\text{Zn}_{(\text{s})} + \text{Cu}^{2+}_{(\text{aq})} \rightarrow \text{Zn}^{2+}_{(\text{aq})} + \text{Cu}_{(\text{s})}$
- b. $2\text{Mg}_{(\text{s})} + \text{O}_{2(\text{g})} \rightarrow 2\text{MgO}_{(\text{s})}$
- c. $\text{HCl}_{(\text{aq})} + \text{NaOH}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$
- d. $\text{Fe}^{2+}_{(\text{aq})} \rightarrow \text{Fe}^{3+}_{(\text{aq})} + \text{e}^-$
- e. $\text{Cl}_{2(\text{g})} + 2\text{e}^- \rightarrow 2\text{Cl}^{-}_{(\text{aq})}$
13. Which set of quantum numbers is not possible?
- a. $n = 2, \ell = 1, m_\ell = 0$
- b. $n = 3, \ell = 2, m_\ell = -2$
- c. $n = 1, \ell = 0, m_\ell = 0$
- d. $n = 3, \ell = 3, m_\ell = -1$
- e. $n = 4, \ell = 1, m_\ell = +1$
14. Which orbital corresponds to $n = 4$ and $\ell = 1$?
- a. 4s
- b. 4p
- c. 4d
- d. 3p
- e. 5p
15. A cell with $E^\circ_{\text{cell}} = -0.45 \text{ V}$ is best described as:
- a. Spontaneous
- b. at equilibrium
- c. Endothermic
- d. Non-spontaneous
- e. Requires external voltage
16. Which of the following is false about a system at equilibrium
- a. $Q=K$
- b. No reactants or products are reacting
- c. [Reactants] and [Product] can vary
- d. Adding a catalyst causes a change in concentrations
- e. There can be more than one phase
17. A mechanism reaction is proposed:





In this mechanism, NO is a...

- a. Reactant
 - b. Product
 - c. An inhibitor
 - d. Catalyst
 - e. Intermediate
18. The Haber process produces ammonia according to the following reaction:
- $$\text{N}_{2(\text{g})} + 3\text{H}_{2(\text{g})} \rightleftharpoons 2\text{NH}_{3(\text{g})} + \text{Heat}$$
- Which of the following changes will shift the equilibrium to produce more ammonia?
- a. Decrease pressure, increase temperature
 - b. Increase pressure, increase temperature
 - c. Increase pressure, decrease temperature
 - d. Decrease pressure, decrease temperature
 - e. Add more ammonia
19. Which of the following will give a solution with the lowest pH if each solution contains a concentration of 0.100M of the given salt?
- a. KNO_2
 - b. NaCl
 - c. K_3PO_4
 - d. HClO_4
 - e. All of these have the same pH
20. A buffer is prepared with a weak acid (HA) and its conjugate base (A^-). If a strong acid (H^+) is added, what happens?
- a. The pH drops significantly.
 - b. The (A^-) reacts with the (H^+) to form (HA)
 - c. The (HA) dissociates into (H^+) and (A^-).
 - d. The buffer capacity is immediately destroyed.
 - e. Nothing happens

1. Answer the following:

Molecules	Electron Counting	Lewis Diagram
SnF₂		
PCl₅⁻		

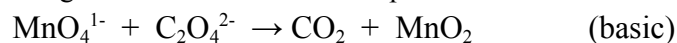
2. Answer the following:

Molecule	Lewis Diagram	Determine molecular geometry (ex. linear)	Determine polarity
ClF₅			
SF₂			

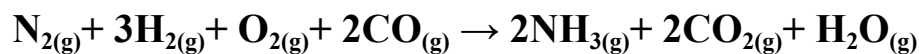
3. Determine the molar solubility of CaF_{2(s)} in a solution containing 0.01000 M NaF, given that $K_{sp} = 1.3 \times 10^{-7}$ at 25°C.

I			
C			
E			

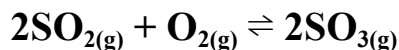
4. Balance the following half-reaction. Show all steps.



5. Consider the reaction:

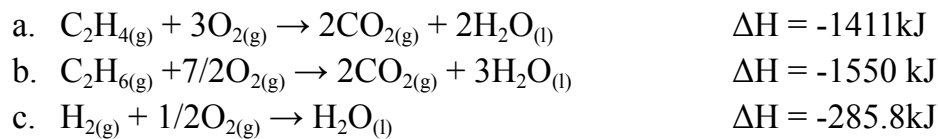


- a. The standard enthalpy change (ΔH) is -4582 kJ at 25°C. Determine the molar enthalpy of ammonia.
6. For the following reaction



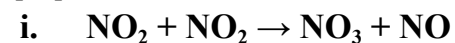
- a. Does this reaction result in a change in entropy, and if so, is ΔS positive or negative?
- b. Calculate the entropy value (ΔS) in J/K given ΔG is -3240.8 kJ/mol at 23°C and $\Delta H = -2291$ kJ:
- c. Under what conditions (if any) would change the spontaneity of this reaction.

7. Given the following data:



Calculate the ΔH for the following reaction: $\text{C}_2\text{H}_{4(g)} + \text{H}_{2(g)} \rightarrow \text{C}_2\text{H}_{6(g)}$. Show all steps!

8. Given the proposed mechanism, answer the following questions.



- What is the overall reaction?
- What are the intermediate(s) in this reaction?
- What are the catalyst(s) in this reaction?
- Explain how you would identify an intermediate vs. a catalyst

8. a) Calculate the enthalpy change ($\Delta H^\circ_{\text{rxn}}$) for the combustion of propane ($\text{C}_3\text{H}_{8(g)}$) using the standard enthalpies of formation provided. $\text{C}_3\text{H}_{8(g)} + 5\text{O}_{2(g)} \rightarrow 3\text{CO}_{2(g)} + 4\text{H}_2\text{O}_{(l)}$

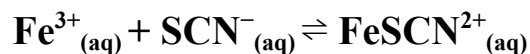
$$\Delta H_f^\circ[\text{C}_3\text{H}_{8(g)}] = -103.8 \text{ kJ/mol} \quad \Delta H_f^\circ[\text{CO}_{2(g)}] = -393.5 \text{ kJ/mol} \quad \Delta H_f^\circ[\text{H}_2\text{O}_{(l)}] = -285.8 \text{ kJ/mol}$$

- b) Write two thermochemical equations for this reaction

- c) Which is a more accurate unit of measurement for energy? Bond energy or heats of formation, explain:

9. A set of standard solutions is prepared by reacting excess Fe^{3+} with known volumes of NaSCN:
8.00 mL of 0.01000 M NaSCN is mixed with 6.00 mL of 0.02000M Fe^{3+} and 6.00 mL of 0.03000 M HCl at 22°C. Because Fe^{3+} is present in large excess, all SCN^- is assumed to react completely to form FeSCN^{2+} .

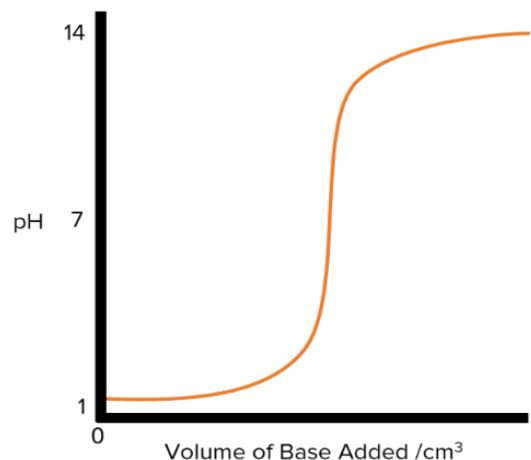
The equilibrium concentration of FeSCN^{2+} is determined experimentally to be: $[\text{FeSCN}^{2+}]_{\text{eq}} = 3.50 \times 10^{-5}$



- a. Determine the equilibrium constant Kc for the reaction:

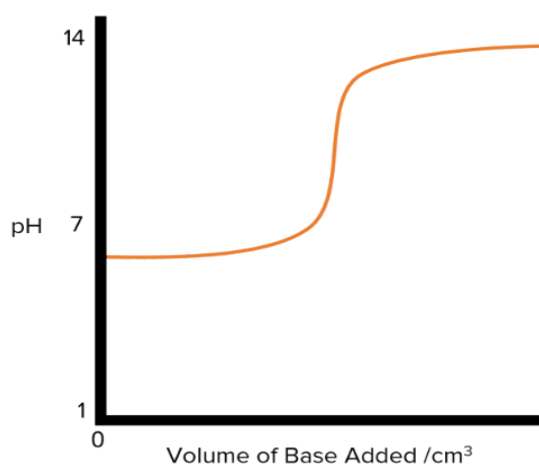
- b. Given Kc at 33°C is 13, is this reaction exothermic or endothermic? Explain.

10. A strong acid-strong base (HNO_3 and NaOH) titration curve and a strong base-weak acid (NaOH and HNO_2) titration curve are shown below. In the area provided, state which acid is being used and provide an explanation.



Acid: _____

Explanation:

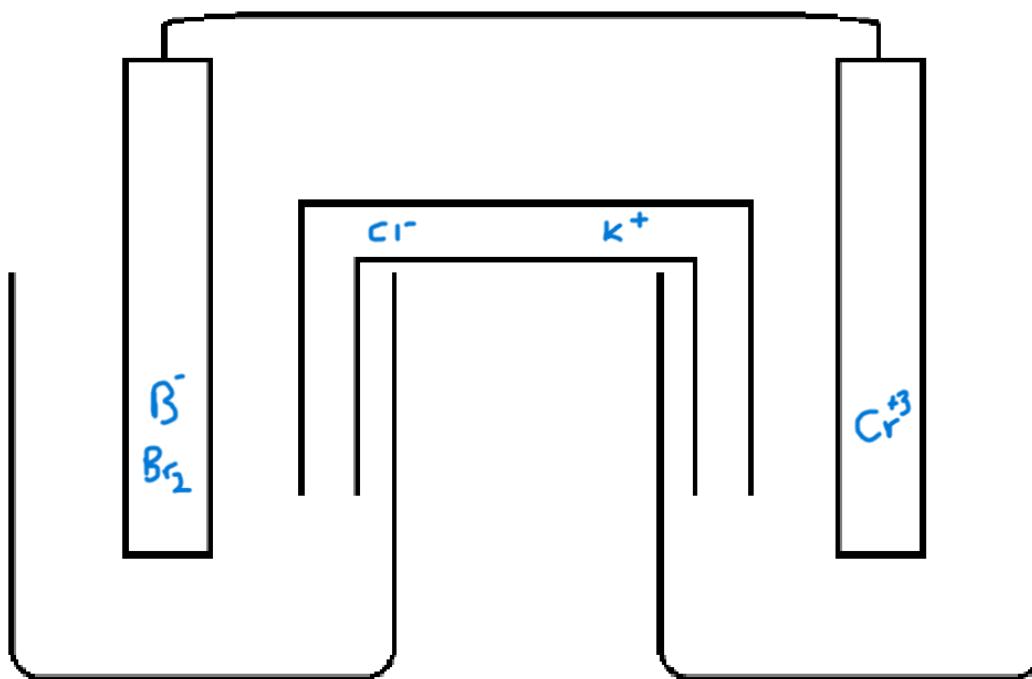


Acid: _____

Explanation:

- a. Why do the pH values differ? Explain using equations.
- b. Which acid has a higher molar concentration? Explain how you know.
- c. On the appropriate graph, label the buffer region and half-equivalence point.

11. A galvanic cell is constructed at 25°C using a $\text{Cr}(\text{NO}_2)_3$ half-cell and a KBr half-cell. The KBr half-cell contains Br^- ions and a platinum (Pt) electrode. The two half-cells are connected by a KCl salt bridge.

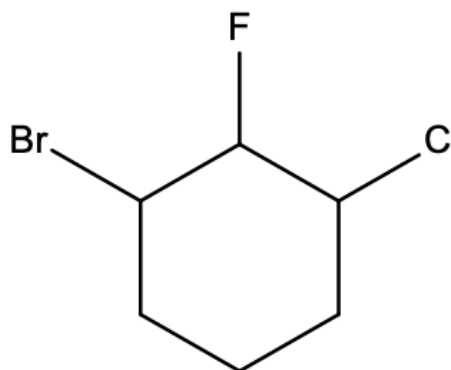
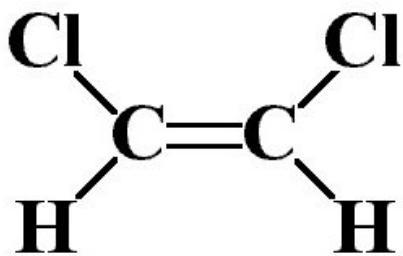


- Label the anode and cathode and show which one is negative/positive.
 - Label the direction of electron flow.
 - Write the half-reactions at the electrodes.
 - Which one is losing mass?
- e. Calculate the E°_{cell}
- f. Explain why a salt solution is necessary for a galvanic cell

12. Answer the following:

Molecule	What crystal is this?	Experiment	Result of experiment
$\text{CH}_3\text{CH}_2\text{CH}_3$		High melting point	
Graphite		Solubility in oil	
Sugar		Conductivity in water	
Magnesium		Hammer Test	

13. Name the following organic molecules:



a) _____ b) _____

14. But-2-ene has two geometric isomers. Draw and name both.